

Current and superfluid fraction for 1D anyons

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Keywords: anyons, quantum matter, ultracold atoms

I. INTRODUCTION

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II. MAIN IDEA

From the previous work with bose gases in one-dimensional lattices [1] they find different physical quantities, from [2]

$$\mathcal{H}^b = -J \sum_{i=1}^M \left(\hat{b}_i^\dagger \hat{b}_{i+1} e^{i\theta \hat{n}_i} + h.c. \right) + \frac{U}{2} \sum_{i=1}^M \hat{n}_i (\hat{n}_i - 1), \quad (1)$$

where $\hat{b}_i^\dagger$ and $\hat{b}_i$ are the creation and annihilation operators of bosons in the lowest Wannier band at site i respectively and $\hat{n}_i = \hat{a}_i^\dagger \hat{a}_i$ is the number occupation operator.

In order to compute the superfluidity of the system, we follow the work for bosons [1] imposing a phase twist ϕ in the entire system for a local unitary transformation to map twisted boundary conditions onto the Hamiltonian [3, 4]. This permits rewrite the Hamiltonian as:

$$\mathcal{H}_\phi^b = -J \sum_{i=1}^M \left(e^{-i\phi/M} \hat{b}_i^\dagger \hat{b}_{i+1} e^{i\theta \hat{n}_i} + h.c. \right) + \frac{U}{2} \sum_{i=1}^M \hat{n}_i (\hat{n}_i - 1), \quad (2)$$

this factor in the hopping term is usually called Peierls phase factor [4]. From the differences of ground energies between the twisted system and the original system we obtain the superfluid fraction [5]

$$f_s = \frac{M^2}{N} \left(\frac{E_\phi - E_0}{J\phi^2} \right), \quad (3)$$

Expanding the terms for the Peierls phase, we obtain:

$$\mathbf{H} \approx \mathbf{H}_0 + \left(\frac{\phi}{M} \right) \mathbf{J} + \left(\frac{\phi^2}{2M^2} \right) \mathbf{T} = \mathbf{H}_0 + \mathbf{H}_{pert}, \quad (4)$$

For the system we defined the following operators:

$$\mathbf{J} = iJ \sum_{i=1}^M \left(e^{-i\theta \hat{n}_i} \hat{b}_{i+1}^\dagger \hat{b}_i - \hat{b}_i^\dagger \hat{b}_{i+1} e^{i\theta \hat{n}_i} \right) \quad (5)$$

$$\mathbf{T} = J \sum_{i=1}^M \left(e^{-i\theta \hat{n}_i} \hat{b}_{i+1}^\dagger \hat{b}_i + \hat{b}_i^\dagger \hat{b}_{i+1} e^{i\theta \hat{n}_i} \right) \quad (6)$$

and we find the superfluid fraction by perturbation theory

$$f_s = \frac{M^2}{NJ\phi^2} \left(\frac{\phi}{M} \langle \Psi_0 | \mathbf{J} | \Psi_0 \rangle - \frac{\phi^2}{2M} \langle \Psi_0 | \mathbf{T} | \Psi_0 \rangle - \sum_{k \neq 0} \frac{1}{E_k^{(0)} - E_0^{(0)}} \right) \quad (7)$$

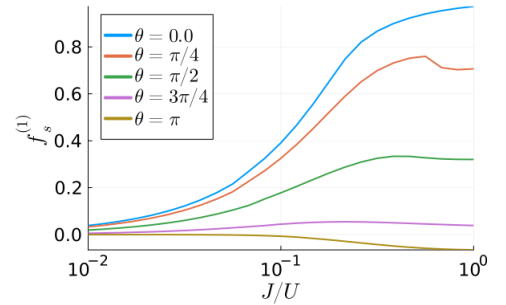


Figure 1. Current $\mathbf{J}$ defined by Burnett for different statistical angles

III. CONCLUSION

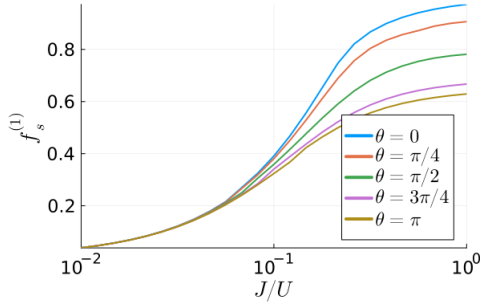


Figure 2. Current $\mathbf{J}$ for anyons

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Appendix: Appendix

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